Mixture of Experts for Tokenformer

(Natural Language Processing Project Proposal)

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Transformers [1] have become the dominant architecture in the field of Natural Language Processing (as well as many other domains), achieving state-of-the-art results. Their computations on a single token are of two distinct types: interactions with other tokens, and with the model parameters. While the first are facilitated by the attention mechanism, which scales greatly, the second type relies heavily on linear projections. This design limits scalability, as size-altering introduces architectural differences, which usually requires retraining the model from scratch (as shown in [2]). This issue has been addressed recently in [3]. The paper proposes to modify the transformer's architecture (achieving, what they call, a *Tokenformer*) by replacing standard linear projections with new *pattention* layers, and treating the model's parameters as tokens, applying a modified attention mechanism on them. This approach significantly improves scalability, while achieving comparable accuracy. These results however beg a question - how does this change influence other modules proposed earlier as transformer modifications? What we want to investigate is this influence on the Mixtral of Experts, introduced in [4]. If the two can work together, we might give rise to new type of transformers, that is both accurate and easily scalable.

The main objective is to implement the Mixtral of Experts (MoE), introduced in [4], but modified with the pattention mechanism described in [3], and then compare it (in terms of both performance and scalability) with two baselines: transformer with MoE, and Tokenformer without MoE (all trained by ourselves with the same setups). In regards to concrete datasets, metrics, and benchmarks, we will follow the Tokenformer paper [3], with some limitations based on our time and compute. That is:

- The dataset will be (a part of) OpenWebText Corpus described in [5].
- We will train smaller models, and try to scale them progressively on small parts of the dataset.
- As the deliverables, we will measure the model's test perplexity against training cost (in TPU minutes / hours), as well as the performance on four widely-recognized zero-shot downstream tasks:
  - Perplexity on the Pile validation split
  - Accuracy on the HellaSwag commonsense reasoning benchmark
  - Accuracy on the PIQA physical reasoning benchmark
  - Accuracy on the WinoGrande pronoun/coreference benchmark

References

[1] Vaswani et al., *Attention is all you need*, 2017

[2] Biderman et al., *Pythia: A suite for analyzing large language models across training and scaling*, 2023

[3] Wang et al., *Tokenformer: rethinking transformer scaling with tokenized model parameters*, 2025

[4] Jiang et al., *Mixtral of Experts*, 2024

[5] Aaron Gokaslan and Vanya Cohen, *Openwebtext corpus*, 2019